NOT MEASUREMENT SENSITIVE

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11 September 1991
SUPERSEDING
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8 November 1985

MILITARY SPECIFICATION

# SEMITRAILERS, TANK: WASTE AND SLUDGE ACID, 3,000 to 5,000 GALLON, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types and three sizes of modified commercial tank semitrailers for collecting, transporting and discharging waste, sludge acids and corrosive liquids weighing up to 13 pounds per gallon. Tank semitrailers procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Classification.

1.2.1 Type. The type of semitrailer shall be determined by its self-contained loading and unloading capabilities and shall be one of the following types, as specified (see 6.2).

Type I - Pressure discharge.
Type II - Pressure discharge, vacuum loading.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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1.2.2 Size. The size of the semitrailer shall be determined by its liquid carrying capacity and shall be one of the following sizes, as specified (see 6.2):

Size A - 3,000 gallon
Size B - 3,600 gallon
Size C - 5,000 gallon

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

| | | |
|---|---|---|
| | W-B-131 | - Battery, Storage: Vehicular, Ignition; Lighting and Starting. |

MILITARY

| | | |
|---|---|---|
| * | MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |

STANDARDS
FEDERAL

| | | |
|---|---|---|
| | FED-STD-297 | - Rustproofing of Commercial (Nontactical) Vehicles. |
| * | FED-STD-595 | - Colors. |

MILITARY

| | | |
|---|---|---|
| | MIL-STD-209 | - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment. |
| | MIL-STD-1223 | - Non-tactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards. |
| * | MIL-STD-1474 | - Noise Limits for Army Materiel. |
| | MS 75021 | - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof. |

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)



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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Hazardous Materials in Tank Motor Vehicles.

(Application for copies of DoT publications should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

| | | |
|---|---|---|
| | J318 | - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted). |
| | J350 | - Spark Arrester Test Procedure for Medium Size Engines (DoD adopted). |
| * | J551 | - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz). |
| | J553 | - Circuit Breakers. |



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| | |
|---|---|
| J560 | - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable. |
| J588 | - Turn Signal Lamps. |
| J682 | - Rear Wheel Splash and Stone Throw Protection (DoD adopted). |
| J700 | - Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DoD adopted). |
| J702 | - Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer. |
| J1292 | - Automobile, Truck. Truck Tractor, Trailer and Motor Coach Wiring. |

(Application for copies of SAE publications should be addressed to SAE, Inc. 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Avenue, Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.6, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All items shall be as represented in the manufacturer's technical data book. Special or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard semitrailer specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.



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* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223, except the exterior color shall be gloss yellow matching color chip No. 13538 of FED-STD-595 and the finish coat shall be with an acid resistant paint.

* 3.1.1.2 Marking and data plates. Identification markings and data plates shall be in accordance with the military requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Instructions and caution plates. Brief operating and precautionary instructions shall be permanently affixed on or near the control panels. The instructions shall be clear, concise and sufficient to enable vehicle use without damage to the equipment or injury to personnel. Sufficient data shall be provided for proper procedure in the preparation of equipment for use. Instruction, data and caution plates shall be engraved phenolic (Formica), 1/8 inch thick, acid resistant or equivalent material.

3.1.1.5 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

* 3.1.1.6 Lifting and tying down attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class l or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings the capacity of each attachment and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

* 3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shal1 comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture .

* 3.2.2 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.



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* 3.2.3 Ratings. The rated payload capacity shall be not less than 39,000 pounds for size A, 46,800 pounds for size B and 65,000 pounds for size C. These weights are computed from the rated volume using 13 pounds per gallon. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity. The axles of the semitrailer shall be positioned so that the proportion of the gross weight of the loaded semitrailer supported on the tires, with a maximum tank slope of 1-1/2 percent to the rear, with the semitrailer on level ground, shall not exceed 32,000 pounds for size A, 34,000 pounds for size B and 46,500 pounds for size C.

* 3.2.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.5 Prohibited materials. Asbestos material shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram of activity per item equals or exceeds 0.01 microuries.

* 3.3 Performance. The semitrailer, fully equipped and loaded, shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unimproved roads (see 6.3.2) and being towed at speeds as great as 55mph over improved roads (see 6.3.1).

3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to a coupled towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

* 3.3.2 Brake performance. The service brakes shall stop the semitrailer-tractor combination, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.3.3 Tracking ability. The semitrailer shall conform to the tracing requirements of Federal Motor Carrier Safety Regulation 393.70(a).

* 3.4 Dimensions and clearances. The semitrailer, without payload, uncoupled from the towing vehicle, resting on its landing legs on level ground, and with the platform level, shall conform to the dimensions specified in table I.



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TABLE I. Dimensions and clearances.

| Feature | Inches |
|---|---|
| Overall length for size A and B, minimum practicable, but not more than (inches) | 480 |
| Overall length for size C, minimum practicable, but not more than (inches) | 510 |
| Overall width not more than (inches) | 96 |
| Overall height for size A and B, not more than (inches) | 132 |
| Overall height for size C, not more than (inches) | 150 |
| Height of upper fifth wheel plate, plus or minus one inch (inches) | 50 |
| Landing gear clearance, radius from center of kingpin to any point 6 inches below upper fifth wheel plate, not less than (inches) | 82 |
| Ground clearance, not less than (inches) | 14 |
| Swing radius from centerline of kingpin to most distant point on the semitrailer nose, not more than (inches) | 54 |

3.5 Frame or frameless type. Chassis construction shall be the manufacturer's standard frame or frameless type, but in either case, the structural integrity requirements of DoT Safety Regulation 178.345 shall be met. Structural members shall support the running gear, upper fifth wheel plate and landing gear.

* 3.5.1 Frame. When a frame type semitrailer is furnished, the frame shall be capable of meeting the performance requirements specified herein. The frame shall be adequately braced to ensure stability. When the vehicle is loaded with specified payload (see 3.2.3), and when being towed by a truck tractor, the maximum fiber stress in any frame member shall not be greater than 33 percent of the frame material yield strength. Crossmembers shall be of full width. Bolsters shall be of sufficient width to support the tank and to distribute the stress when the tank is loaded with specified payload. All parts of the frame shall be completely electric arc welded.

3.5.2 Frame splices. Where frame members are spliced, joints shall be reinforced to fulfill the dynamic loading requirements and the static loading requirements. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

* 3.5.3 Stress analysis. A stress analysis for the semitrailer, loaded as specified in 3.4, shall be performed. The stress analysis shall include shear and moment diagrams and deflection calculations. This analysis shall verify compliance of the chassis and tank structural integrity with Safety Regulation 178.345.3 and shall at least the following calculations and diagrams:



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(a) The sprung GVW load distribution diagrams
(b) The vertical shear diagram for the length of the vehicle
(c) The bending moment diagram for the length of the vehicle
(d) The loads, sections, shear, and bending members that carry maximum shear and moment, to include the analysis of the stresses in attaching weldments or other fastenings
(e) The loads, sections, shear, and bending moment diagrams at kingpin and rear under structure subframe members, and for the drop section when drop frame semitrailers are provided
(f) The load, section, shear, and bending moment diagrams for the most critical crossmember, to include the analysis of stresses in attaching weldments or other fastenings
(g) The loads, sections, and stresses in members that carry maximum compression
(h) Consider all potential structural loadings
(i) List all assumptions and references utilized to provide required data.

3.6 Suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, when the semitrailer is loaded with its rated payload (see 3.2.3). Clearances shall be sufficient to prevent interference between tires and any other part of the semitrailer under the operating conditions specified herein.

3.7 Tandem axle. The semitrailer shall be furnished with tandem axles. Axle ratings shall be at least equal to the load imposed on each axle measured at the ground, with the semitrailer loaded with rated payload as specified in 3.2.3. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for replenishing the oil supply. The hub caps shall incorporate provisions for venting or equivalent method of withstanding internal pressure buildup without leakage.

* 3.8 Wheels, rims, tires, and tubes. The semitrailer shall be equipped with dual wheels, rims and tires. The rim and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Rim sizes shall be the same for all wheels on the semitrailer. Tires size and load range (ply rating) shall be the same for all tires on the semitrailer. Multi-piece rims shall not be furnished. When specified (see 6.2), disc type wheels shall be furnished.

* 3.8.1 Tires. Tires shall be tubeless type with highway tread. Tires shall be of the steel belted radial type. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded with rated payload. Tires



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shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.8.2 Carrier for space tire assembly. Unless otherwise specified (see 6.2), a. carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.19. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.8.4 Spare wheel. When specified (see 6.2), a spare wheel or rim shall be mounted in the tire carrier. When a spare tire assembly is specified (see 3.8.5), a spare wheel or rim shall be furnished.

3.8.5. Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim specified in 3.8.4. The spare tire shall be of the same size and tread design and of the same load range (ply rating) as tires installed on the semitrailer.

3.9 Brakes.

* 3.9.1 Brakes. The brakes shall be of the full air internal expanding type and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43,393.45,393.46, and 393.47. Brake linings shall be of nonasbestos material. The brake system shall include automatic slack adjusters, piping, hose connections gladhands, spring loaded dust covers or dummy gladhands equipped with security chains or cables, air reservoirs with remote control drain, brake chambers, relay emergency valve, and all other parts and connections required for a complete air brake system. Gladhands shall conform to SAE J318. The location of the gladhands shall be as recommended in SAE J702. All components of the brake system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier. No part of the brake system shall extend below the bottom of wheel rims to insure, in case of a flat tire, that the weight of the semitrailer will be supported by the rim and the flat tire and not be imposed on any component of the braking system.

3.9.2 Parking brake. Spring or air diaphragm mechanical lock type parking brakes shall be provided. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency braking conditions. The parking brakes shall remain in



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the applied conditions with no additional application and despite the depletion of all air pressure. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.10 Upper fifth wheel. The upper fifth wheel plate shall completely cover a 36 inch diameter lower fifth wheel when the lower fifth wheel is coupled to the kingpin of the upper fifth wheel and the semitrailer is turned at all angles. The upper fifth wheel shall conform to Federal Motor Carrier Safety Regulation 399.70(b).

* 3.10.1 Upper fifth wheel plate. The upper fifth wheel plate shall be not less than 0.25 inch thick. The plate shall be reinforced on the upper surface through the use of commercial steel angles, channels, and ribs. The plate shall include an access hole to permit servicing the brake connections and electrical connectors when these components are not serviceable from the top.

* 3.10.2 Pickup plate. A turned up lip pickup plate, compatible with the size (payload) of the semitrailer, shall be provided as an extension of the upper fifth wheel plate. The pickup plate shall have a turned up lip not less than 2 inches high. The pickup plate shall extend sufficiently forward to protect the semi trailer from damage during coupling and uncoupling operations to and from a truck tractor. The pickup plate shall be integral with the upper fifth wheel plate.

* 3.10.3 Kingpin. A 2-inch diameter kingpin conforming to SAE J700 shall be furnished. The kingpin shall be fabricated of heat treated alloy steel.

* 3.11 Lighting. The lighting system shall be of 12-volt potential and shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25, and 393.26. The brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected) or semitrailer bumper. The front of the semitrailer shall be equipped with a receptacle conforming to SAE J560 with the conductors connected and color coded as specified therein. The receptacle shall be located in accordance with SAE J702. Clearance lights shall be constructed for easy removal and replacement of lamps and lenses without the use of hand tools. Turn signal lamps shall conform to SAE J588.

3.11.1 Wiring. Wiring shall conform to SAE J1292, type Heavy Duty Thermoplastic (HDT) or Heavy Duty Rubber Single Braid (HDB), and to Federal Motor Carrier Safety Regulations 393.27 through 393.30, 393.32 and 393.33. All wiring shall be firmly attached to the semitrailer. Wiring shall be in rubber covered cable and individual lines shall be run in plastic tubing or metal conduit. Junction boxes shall be of vaporproof design.



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3.11.2 Circuit protection. The electrical system shall have circuit breakers. The circuit breakers shall conform to SAE J553 and shall be located near the wiring connector socket, readily accessible for service. Switches and circuit protection devices installed on the semitrailer shall be of permanent vaporproof design or enclosed in vaporproof boxes, except circuit breakers located at the front of the semitrailer may be installed in a weatherproof box away from tank vents or openings.

3.11.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system independent of the 12-volt dc lighting system shall be provided for use with a towing vehicle equipped with a nominal 24-volt dc system. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.11 through 3.11.3. The 24-volt dc system shall conform to the Federal Motor Carrier Safety Regulations referenced herein, except the 24-volt dc stop and turn lamps shall be combined.

3.11.4 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt dc system (see 3.11) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.11.5 Receptacle 24-volt dc. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

| | |
|---|---|
| Contact B. | Connect to left-hand turn signal and stop lamp (yellow). |
| Contact D. | Connect to ground (white). |
| Contact E. | Connect to clearance, side marker, identification and tail lamps (black and brown). |
| Contact J. | Connect to right-hand turn signal and stop lamp (green). |
| Contact L. | Connect to ground (white). |

The remaining contacts shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.11.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.11.4), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 28-volts dc to 12-volts dc. Each circuit



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resistor shall be selected to reduce the 28-volt dc regulated voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing, shall be provided with adequate ventilation or a heat sink to prevent overheating and any damage to resistors, wiring, or adjacent components, and shall not be located in close proximity to any vent or pump exhaust.

3.12 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating landing legs with two speed gears and a handcrank on the curb side. Landing legs shall be equipped with steel wheels or self-leveling skid pads. Both legs shall be capable of being simultaneously operated by a handcrank from the curbside. Supports for the crank extension shafts and clips for holding the crank when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency.

3.12.1 Landing gear range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel from 47 inches to not less than 52 inches from the ground. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall be not less than 14 inches.

3.13 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regtuation 393.86, and shall conform to DoT Safety Regulation 178.340.

* 3.14 Rear wheel splash and stone throw protection. Real wheels shall have fender extensions or mud flaps at the rear. Splash and stone throw protection shall be in accordance with SAE J682. The semitrailer shall have skirting or fenders that cover the area above the wheels. The skirting on the fenders shall be essentially flat on top and capable of supporting a person weighing 250 pounds. Fenders shall be aluminum or stainless steel. A step and grab handle for ease of mounting the top of the curbside fender shall be provided. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be used as a backing to prevent bolt head or bolt nut from rupturing the mud flap during tightening operation upon installation. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.25 inch wide by 0.094 inch thick shall be furnished at each bolt.

3.15 Lubrication. Lubrication means shall be provided for all part of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used.



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* 3.16 Tank body and equipment, type I and type II. The semitrailer shall be designed and constructed for the legal transportation by interstate carriers of liquid corrosive materials of the types specified (see 6.2). Accordingly, the semitrailer shall conform to the regulations of the Department of Transportation (DoT) governing the transportation of the specified corrosive materials in tank motor vehicles. Conformance to DoT Safety Regulation 178.345 and DoT Safety Regulation 178.348, Specification DoT-412, as published in the latest issues of the Code of Federal Regulations and as modified by all issues of the Federal Register published and in effect prior to completion of the semitrailer shall be total and without exception or exemption. Tank marking shall be in accordance with all the applicable portions of DoT Safety Regulations 171 through 178 for hazardous material.

* 3.16.1 Material. The tank and all its associated components, saddles, pads and appurtenances shall be fabricated of type 316L stainless steel, suitable for handling liquids of the types specified (see 6.2). The tank and the lining, when furnished, shall meet the requirements of DoT Safety Regulation 178.345-2, without repair or replacement, with a thinning by corrosion allowance sufficient for a 10 year normal service life of the vehicle. The top of the fenders, the flashing boxes, and the exterior tank discharge and intake pipe areas shall be of type 316L stainless steel. Gaskets and seals shall be of suitable material to withstand attack by the tank contents. When specified (see 6.2), in addition, all tank surfaces exposed to the specified liquids during loading, unloading and transportation shall be completely covered with natural rubber lining, synthetic rubber ("Hypalon") lining or polyester (type 64) lining, as appropriate to the type of liquids specified. The lining material shall be not less elastic than the material of the tank proper and shall be leakproof. When lining is furnished the lining material shall also be applied to the exterior tank section at the flashing boxes and on top of the rear fenders.

3.16.2 Capacity. The liquid capacity of the tank shall conform to 1.2.2 for the specified size of vehicle, plus at least 3 percent for expansion. The tank shall have a single compartment.

3.16.2.1 Capacity marker. A graduated capacity marker or indicator of corrosion resistant construction shall be installed in a location readily visible from the operator's normal position during loading operations.

* 3.16.3 Tank configuration. The tank shall be cylindrical or round in cross section with external ring stiffeners.

3.16.4 Design pressure. The tank shall be designed and constructed to be unloaded by an internal pressure of not less than 35 pounds per square inch gage (psig).

3.16.5 Product density. The tank shall be designed and constructed to transport liquids with a density of up to 13 pounds per gallon.



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* 3.16.6 Baffles. Unless a lined tank is furnished, the tank shall be furnished with interior baffles. Not less than two baffles shall be provided. Baffles shall be flanged and dished. Half or full round openings, top bottom, of not less than 3.3375 inch minimum radius shall be provided in each baffle. Equivalent area openings shall be provided at the 3 o'clock and 9 o'clock positions of the baffles. Each baffle shall also be provided with not less than an 18 inch diameter manway opening. The edges of all baffle openings shall have a flange of approximately 1 inch.

* 3.16.7 Rear manhole. The top rear end of the tank shall be provided with a manhole opening not less than 18 inches in diameter. A light weight, liquid tight, easy opening, locking manhole cover or a bolted cover with a liquid tight easy opening, locking fill cover not less than 10 inches in diameter shall be supplied. The fill opening shall be provided with a removable screen basket not more than 36 inches deep. The screen basket screen openings shall be not more than 0.50 inch square. The screen basket shall be utilized to catch rags or other solid waste when loading the tank. Hinged covers shall be hinged at the forward edge.

3.16.8 Front manhole. When specified (see 6.2), the top front end of the tank shall be provided with a second manhole, identical to that specified in 3.16.7, except that the fill opening and basket screen are not required.

3.16.9 Flashing boxes. Combination flashing and overturn protection boxes shall be provided around each manhole. Each box shall be provided with two drain lines, discharging toward the ground under the tank.

* 3.16.10 Platforms and ladders. A platform approximately 30 inches below the flashing box shall be provided on the curbside, extending from the front to the rear flashing box. The platform shall be fabricated from non-skid metal grating not less than 18 inches wide. The platform shall be provided with guard rails. Steel rung ladders with handholds shall extend from the top of the tank to the platform and down to the top of the curbside fender (rear ladder) or bottom of the tank (front ladder)

3.16.11 Top cleanouts. At the top front of the tank and at the top rear of each baffle, provisions for inserting cleanout tools shall be provided. The cleanouts shall be 3 inch nipples with blank, bolted on flanges. A front cleanout is not required when front manhole is furnished.

* 3.16.12 Safety vents. The safety vents required by DoT Safety Regulation 178.348, Specification DOT-412, shall be installed at the appropriate longitudinal location on the tank. A cover hose assembly to direct the safety vent discharge toward the ground, under the tank, shall be provided.

* 3.16.13 Tank Support. Support of tank shall conform to DoT Safety Regulation 178.345-6. Tank semitrailers provided with continuous external frame type construction shall have supports directly under each bulkhead, baffle, and at the front and rear of the tank. Adequate support



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required by DoT Safety Regulation 178.3456(b) shall be provided when frameless type tank construction is used in which the tank constitutes in whole or in part the structural member in lieu of a frame.

3.17 Type I tank body and equipment. In addition to the requirements of 3.16 through 3.16.13, type I semitrailer shall be furnished with the additional equipment specified in 3.17.1 through 3.17.6.

3.17.1 Sump and bottom cleanout. A 6-inch minimum diameter sump with a bolted flange closure plate shall be provided at the bottom rear of the tank.

3.17.2 Dip pipe. A pipe of type 316L stainless steel extending from within the 6-inch sump to the outside of the tank at the top rear, within the rear flashing box, shall be provided. The pipe shall be flanged to the top of the tank and shall be not less than 3 inches in nominal diameter.

* 3.17.3 Outlet valves and piping. The top of the dip pipe shall be provided with a 3-inch nominal diameter discharge valve with the valve control readily accessible from the rear platform. Automatic means of remote closure conforming to DoT Safety Regulation 178.345-9 shall be included. Piping of type 316L stainless steel not less than 3 inches in nominal diameter shall extend down from the valve to a convenient working height from the ground at the rear and shall terminate with a second 3-inch valve and a blank, bolted flange, complete with security chain.

3.17.4 Air inlet assembly. An air inlet assembly shall be provided at the rear manhole for discharging the payload using the self-contained air pressure system (see 3.17.5). The air inlet line shall be at least a one inch diameter pipe. The pipe at the air compressor unit shall be provided with an air pressure gage visible to the operating personnel. The pipe leading from the compressor shall be provided with a 35 psig pressure relief valve, a check valve, a tee with a drain line to the ground ending with a drain cock, and terminate with an air inlet fitting. A rubber hose of sufficient diameter and length with couplings (Dickson-Boss type or equivalent), shall be furnished for installation between compressor line and tank line.

3.17.5 Pressure system. A self-contained pressure system for unloading the tank shall be furnished. The system shall include a diesel engine driven air compressor with a capacity of not less than 100 cubic feet per minute (cfm). The compressor shall be a two stage air cooled type. The air compressor shall be mounted in conjunction with the engine (see 3.17.5.1), with mounts of the type that shall minimize transfer of the compressor engine vibration to the semitrailer. The compressor engine unit shall be provided with a suitable platform and housing. Sufficient access for personnel to operate and maintain the unit shall be provided within the housing area.



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* 3.17.5.1 Engine. The compressor drive shall be a diesel engine. The engine shall be an air cooled type, of standard commercial design with the power, torque and speed necessary to satisfactorily operate the air compressor. The engine shall be furnished complete with all accessories necessary for operation, including air cleaner, starting system, alternator or generator, battery fuel tank of sufficient capacity for eight hours of normal operation, exhaust system (see 3.17.5.2), engine mounted instrument panel with throttle control, start/stop switch and battery charging indicator. The battery shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. The engine shall be provided with an ether aid cold start system, glow plug or grid heater.

3.17.5.2 Engine exhaust. system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83, and unless otherwise specified (see 6.2), shall be furnished with spark arrester. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. Means shall be provided to vent exhaust fumes away from payload discharge outlet(s) and operating personnel.

* 3.17.5.3 Radio interference suppression. The engine shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.17.5.4 Engine sound level. The engine noise level during operation shall be not more than 84 db(A) when measured in accordance with MIL-STD-1474. The noise level shall be measured at the operator's station adjacent to the controls with the operator standing on the ground.

* 3.17.6 Hose and hose carriers. Unless otherwise specified (see 6.2), 3-inch inside diameter (id) discharge hose, fabricated of material to withstand the acids specified, shall be furnished. Four 25-foot lengths of such smooth bore discharge hose, rated for not less than 100 psig, shall be furnished. Each hose section shall be fitted with male and female couplings of the cam-lock type, rated for not less than 100 psig. The hose fittings shall be of stainless steel or plastic, as needed to withstand the acids specified. Troughs, with drainage, shall be provided for stowing the hose sections. The hose sections shall be attached to the troughs with hose clamps. The troughs and hose clamps shall be fabricated of the same material as necessary for the cam lock couplings.

3.18 Type II tank body and equipment. In addition to the requirements of 3.16 through 3.16.13, type II semitrailers shall be furnished with the additional equipment specified in 3.18.1 through 3.18.11.

3.18.1 Liquid collection equipment. Liquid contents shall be drawn into the tank through a hose line by suction provided by a partial vacuum developed within the tank. Unloading of



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liquid from the tank shall be by gravity and by compressed air to empty the tank of liquid through a rear gate valve. The required partial vacuum and air pressure shall be generated by a dual purpose pump furnished with the semitrailer and designated herein as the vacuum pump. The vacuum pump shall be driven by an auxiliary diesel engine.

* 3.18.2 Liquid intake. A 3-inch nominal diameter, stainless steel, quick opening gate valve for liquid intake shall be provided. Automatic means of closure conforming to DoT Safety Regulation 178.345-6 shall be included. Liquid intake shall be through the valve at the rear top of the tank, or at the rear bottom through a vertical standpipe, inside the tank. The standpipe shall extend within 18 inches of the tank top and shall be provided with a baffle wear plate. The standpipe shall be not less than 3 inches in inside diameter. The 3-inch valve shall be equipped with a cam-lock coupler and a dust cap with security chain, all of stainless steel.

* 3.18.3 Liquid discharge. A 4-inch nominal diameter, stainless steel, quick opening gate valve for liquid discharge shall be provided. The valve shall be at the bottom, rear of the tank. The valve shall be equipped with a cam lock coupler and a dust cap with security chain or cable, all of stainless steel. The valve remote control, as required by DoT Safety Regulation 178.348. DoT-412, shall be located at the front, streetside of the semitrailer. The control shall be outlined by a contrasting color panel at least 12 inches square, identified with the words "EMERGENCY SHUTOFF", and provided with indicator arrows or instructions such as "PUSH", "PULL" or other appropriate wording. Identification and instruction wording shall be in letters not less than 2 inches high.

3.18.4 Tank and vacuum pump performance. The vacuum pump shall provide sufficient suction to cause an average water flow rate into the tank of not less than 200 gpm with a vertical lift of 10 feet. Evacuation of liquid from the tank shall be by pressure sufficient to cause an average water flow rate of not less than 150 gpm against a head of 15 feet, when using compressed air from the discharge side of the vacuum pump. All inlet and outlet piping shall be capable of passing solids up to 2-inch sphere size. Float shutoff valves shall be provided in the liquid tank and the surge tank, capable of closing off the vacuum line, when under suction from the vacuum pump, without transmitting any liquid to the vacuum pump when either the liquid tank or the surge tank is filled with liquid. The tank liquid float valve and surge tank shall be capable of preventing foam from entering the vacuum pump suction line when the semitrailer is loading waste caustics and detergents, to within an allowable leakage tolerance of 0.02 percent of the tank capacity collected into the tank.

* 3.18.5 Piping. All vacuum lines inside and outside the tank shall be of type 316L stainless steel. The pipe and fittings shall be not less than 2.50 inch standard pipe size and shall be made up in subassemblies and interconnected by boiler unions or similar quick disconnects. All liquid suction and discharge piping shall be of type 316L stainless steel. The pipe and fittings shall be



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not less than 3-inch standard pipe size. All valve handles, of which one shall be provided for each valve, shall be accessible from the ground. The piping system shall withstand all expected expansion and vibration for the normal service life of the vehicle.

* 3.18.6 Vacuum pump. The vacuum pump shall be a water or air cooled, multicylinder, reciprocating piston pump or rotary type pump with positive displacement lubricating oil pump, capable of the performance required in 3.18.4 for both partial vacuum and compressed air output, and shall be furnished complete with all necessary accessories. The pump shall be rated for continuous duty. Vacuum pump mounting shall be such as to prevent undue vibration. Ground clearance for the vacuum pump shall be not less than 14 inches. A four-way valve, or four valves, shall be provided on vacuum pump manifold in order to permit the tank supply line to be switched between the partial vacuum and the compressed air outlets of the vacuum pump. Complete manual control of the four-way valve, or the four valves, shall be furnished at the valve or valves, and, through mechanical linkage to the side, shall be operable by an operator from ground level. Valve(s) shall be operated from a control lever capable of remaining in any position setting unaffected by vibration. The pump suction and discharge openings shall be a minimum of 2.50 inches id. All areas of the vacuum pump which may come into contact with vapors shall be resistant to the vapors of the liquids specified. The pump shall have a continuous duty capacity of 150 cfm at 26 inches of mercury vacuum, and a minimum discharge pressure of 35 psig.

3.18.7 Surge tank. A surge tank of not less than l-gallon capacity shall be furnished in the vacuum line between the tank and vacuum pump. Means shall be provided to drain and clean the surge tank.

* 3.18.8 Pump engine. The pump shall be driven by a diesel engine. The engine shall be of standard commercial design with horsepower, torque and speed necessary to satisfactorily operate the pump for continuous maximum performance requirements of 3.18.4. The engine shall be mounted on a structural sub-base with vibration dampening devices. The engine shall be furnished complete with all accessories necessary for operation, including air cleaner, a complete 12-volt starting system, alternator or generator, battery, a fuel tank of sufficient capacity for eight hours of normal operation, sediment bowl or fuel filters, an engine mounted instrument panel with throttle control, start/stop switch, battery charging indicator and an exhaust system (see 3.18.9). The battery shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. The engine shall be provided with an ether aid cold start system, glow plug or grid heater. Provisions shall be made to start the engine without vacuum pump load. When "V" belts are furnished the rating of "V" belts shall be not less than 150 percent of the power transmitted.

3.18.9 Engine exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83, and, unless otherwise specified (see 6.2), shall be furnished with spark



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arrester. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350. Means shall be provided to vent exhaust fumes away from payload discharge outlet(s) and operating personnel.

* 3.18.9.1 Radio interference suppression. The engine shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.18.9.2 Engine sound level. The engine noise level during operation shall be not more than 84 db(A) when measured in accordance with MIL-STD-1474. The noise level shall be measured at the operator's station adjacent to the controls with the operator standing on the ground.

3.18.10 Hoses and hose carriers. Unless otherwise specified (see 6.2), the manufacturer's standard 3-inch id, noncollapsible suction hose, and 4-inch id discharge hose shall be furnished. Hoses shall be smooth bore, rated at full vacuum and pressure of not less than 100 psig. Each hose section shall be fitted with male and female couplings of the cam-lock type, rated for not less than 100 psig. The hose fittings shall be of stainless steel or plastic, as needed to withstand the acids specified. A total of eight, 25-foot lengths of suction hose and four, 25-foot lengths of discharge hose shall be furnished. Troughs, which shall have sufficient drainage, shall be provided for stowing the hose sections. Hose sections shall be attached to the troughs with hose clamps. The troughs and hose clamps shall either be fabricated of the same material as necessary for the hose fittings.

3.18.11 Gage(s). Gage(s), with not less than 4-inch diameter dial(s) to indicate the partial vacuum attained when loading tank and the air pressure during, the time tank is being unloaded, shall be furnished.

3.19 Tools. When specified (see 6.2), the semitrailer shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a jack, handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.19.1 Tool Stowage. Storage space of sufficient size to accommodate vehicle jack, wheel nut wrench and hand tools shall be furnished for retaining equipment during vehicle operation. Storage space for these tools shall be located under the vehicle frame, shall be weatherproof and shall provide for locking with a padlock.

* 3.20 Servicing and adjustment. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for operational use, including at least the following: proper adjustment of electrical and brake systems, inflation of all tires, servicing of liquid cooled engine cooling systems with a solution of ethylene glycol type



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antifreeze and water in equal parts by volume, and complete lubrication of chassis and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.21 Workmanship. Defective components, or parts and assemblies which have been repaired or modified to overcome deficiencies, shall not be furnished. All construction techniques utilized in the tank semitrailer manufacture shall be in accordance with the highest standards of the industry. All materials used shall be prime quality.

3.21.1 Welding. Proper preparation of metal prior to welding shall be provided. Surface of parts to be welded shall be free of rust scale, paint grease and other foreign matter. Welding rods shall be compatible with the metal being used. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture.

## 4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.



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4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction, and in the presence, of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine net weight and distribution of net weight on kingpin and rear axles. The imposed loading on kingpin and rear axles will be computed using the net weight with the rated payload (see 3.2.3) in the tank. The calculated imposed load on rear axles will be utilized to ascertain that the suspension, axles and tires are of adequate capacity to meet contract requirements.

4.3.2 Road test. The road test shall consist of coupling the semitrailer, loaded with water, to a truck tractor. The tractor-trailer combination shall be driven a distance of not less than 100 miles under the conditions specified in 3.3. At least 20 percent of the distance shall be over unimproved roads. At least five sudden stops shall be made from a speed of not less than 20 mph with tank filled to approximately 60 percent capacity, to determine that baffles are of adequate structural strength. During the road test, the semitrailer shall be observed to determine conformance to tracking ability requirements specified in 3.3.3.

* 4.3.2.1 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a. major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting



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(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

* 4.3.3 Leakage test. After the road test, the completely assembled tank-body, including all piping, shall be tested for leaks in accordance with DoT Safety Regulation 178.348, Specification DoT 412. In addition, all piping and hoses shall be tested for leaks at the maximum pressure and vacuum they can encounter during use of the vehicle mounted pumping system.

4.3.4 Functional tests. The pressure discharge and vacuum loading (type II only) systems shall be cycled through all functions, using water. All valves shall be checked for proper operation.

4.3.5 Electrical test. An electrical test shall be conducted on the first production vehicles to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test the resistor assembly, including resistors, wiring and adjacent components, shall be examined for evidence of overheating, deterioration or damage.

* 4.3.6 Availability of regulations. The contractor shall make available to the Government at the point of final acceptance a current copy of DoT Safety Regulation178.345 and DoT Safety regulation 178.348, Specification DoT-412. The document shall be surrendered to the Government for the personal use of the Government representatives in conjunction with the contract for the duration of the contract.

* 4.3.7 Joint specimens. The contractor shall offer for Government inspection and approval the joint test specimens subjected to the tensile test required by DoT Safety regulation 178.345 together with certification regarding their date of preparation and testing. The test specimens shall be current in accordance with 178.345.

* 4.3.8 Certification. A manufacturer's certificate of compliance to DoT-412 as defined in 178.345 shall be furnished and securely stowed and shipped with the vehicle together with instructions for the recipient regarding the certificate's retention and use. The certificate shall bear, in addition to the signatures specified above, the typed and signed name of the Government official accepting the vehicle.



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4.3.9 Production sample. Upon acceptance the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.10 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance; records acceptable, to the Government indicating that the servicing and adjusting required by 3.20 has been accomplished.

## 5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

## 6. NOTES

6.1 Intended use. The semitrailers are primarily intended for, collecting and hauling waste and sludge acids and corrosive liquids weighing up to 13 pounds per gallon to disposal sites.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title number and date of this specification.
(b) Type of semitailer required (see 1.2.1).
(c) Size of semitrailer required (see 1.2.2).
* (d) Issue of DODISS to be cited in the so1icitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) Identification of the appropriate services for treatment and painting (see 3.1.1.1).
(f) Identification of the appropriate service for making and data plates (see 3.1.1.2).
(g) Rustproofing, if required (see 3.1.1.3).
(h) Tropical rustproofing, if required (see 3.1.1.3).
(i) Lifting attachments, if required (see 3.1.1.6).
(j) Tiedown devices if required (see 3.1.1.7).
(k) Disc type wheels, if required (see 3.8).
(l) Spare tire carrier, if not required (see 3.8.2).
(m) If spare wheel or rim is required (see 3.8.4).



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(n) A spare tire, if required (see 3.8.5).
(o) An independent 24-volt system, if required (see 3.11.3).
(p) An interconnected 24-volt system, if required (see 3.11.4).
(q) Type(s) of liquid(s) to be transported, the pH value of the liquid(s), and the diameter of the solids present in the liquid(s) (see 3.16 and 3.16.1).
(r) A lined tank, if required (see 3.16.1).
(s) A front manhole, if required (see 3.16.8).
(t) If hoses and hose carriers are not required for type I (see 3.17.6).
(u) If spark arrester is not required (see 3.17.5.2 and 3.18.9).
(v) If hoses and hose carriers are not required for type II (see 3.18.10).
(w) If tools are required (see 3.19).

6.3 Definitions.

6.3.1 Improved road. An improved road is smooth, hard surface road, such as a concrete or asphalt paved highway.

6.3.2 Unimproved road. An unimproved road is an unpaved unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

* 6.4 Subject term (key word) listing.

Acids
Baffles
Landing gear
Manhole
Pressure discharge
Vacuum loading.

6.5 Changes from previous issue. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections. deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodians:
Army - AT
Navy - YD
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